Rhythmic Segment Analysis: Supplementary materials

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Abstract

This document contains supplementary materials.

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1 Backmatter

1.1 Data and code

The Python package rhythmic-segments can be installed using pip, and documentation is available on bacor.github.io/rhythmic-segments. Data and code used for the examples in this paper is available online: **[will be deposited to Zotero upon acceptance]**. Note in particular that it contains a small LaTeX package used to produce figures 2 and 3.

1.2 Use of AI tools

OpenAI's Codex model assisted the development some of the research code, but always instructed and supervised by the author: much of the Python package was for example written in collaboration with Codex, in particular the documentation and test suites. ChatGPT was used when brainstorming terminology and presentation of the ideas in this paper: the term 'quantal' was for example suggested by ChatGPT. The first draft of this paper was written solely by the author and is available upon request. Afterwards, spelling and grammar were corrected using feedback from a chatbot; editorial suggestions for improved fluency were also included.

2 Mathematical framework

This section develops the main concepts underlying rhythmic segment analysis in a more formal fashion.

2.1 Segments and patterns

Consider a dataset $\{t_1, \ldots, t_{K+1}\}$ of distinct events ordered in time, and let $i_k = t_{k+1} - t_k$ denote the time intervals between them. We may in practice assume that intervals are strictly positive, but formally we can allow for zero intervals. Any sequence of n consecutive (and nonnegative) intervals $\mathbf{x} = (i_k, \ldots, i_{k+n})$ will be called a *segment* of length n. We will use the the L_1- norm,

$$\|\mathbf{x}\|_1 = |x_1| + \cdots + |x_n|, \tag{1}$$

so that the norm or length of a segment equals its duration. Accordingly, we measure the distance between two segments $\mathbf{x}$ and $\mathbf{y}$ as

$$\|x - y\|_1 = \sum_{i=1}^{n} |x_i - y_i| \tag{2}$$

This distance is known as the *taxicab* distance, but we will refer to it as the *segment distance* d_s.

Definition 1. The *segment space* $\mathcal{S}^n$, for $n \geq 2$, consists of non-zero and non-negative vectors $\mathbb{R}^n_{\geq 0} \setminus \{0\}$ equipped with the L_1-norm $\|\cdot\|_1$ and the *segment distance* $d_s(x, y) = \|x - y\|_1$.

Two segments $(0.1, 0.2, 0.1)$ and $(2, 4, 2)$ can have a different duration, but still form the same rhythmical *pattern*. That pattern can be expressed equivalently by either the *ratios* $1 : 2 : 1$ or by a normalized vector $(0.25, 0.5, 0.25)$ containing the relative durations of all intervals. The $(n-1)$-dimensional *simplex* Δ^{n-1} is exactly the space of nonnegative and normalized vectors:

$$\Delta^{n-1} = \{\mathbf{x} \in \mathbb{R}^n_{\geq 0} : x_1 + \ldots x_n = 1\}. \tag{3}$$

We will equip the simplex with the *total variation distance*, which is half the taxicab distance, and will refer to it as the *pattern distance* d_p. This is distance is normalized, so that the total variation distance between any two corners of the simplex is 1.[1]

Definition 2. For $n \geq 2$, the *pattern space* $\mathcal{P}^n$ is the simplex Δ^{n-1} equipped with the *total variation distance* $d_p(\mathbf{p}, \mathbf{q}) = \frac{1}{2}\|\mathbf{p} - \mathbf{q}\|_1$.

2.2 Pattern-duration space

Every segment $\mathbf{x}$ can be uniquely decomposed into a pattern $\mathbf{p} = \mathbf{x}/\|\mathbf{x}\|_1$ and a duration $d = \|\mathbf{x}\|_1$. Figure 1 illustrates that you can think of this as a coordinate transform to the point (p, d) in the *pattern-duration space* $\Delta^{n-1} \times \mathbb{R}_{>0}$.

[1] Two corners $\mathbf{p}$ and $\mathbf{q}$ have all-zero coordinates, except for a single 1 in different places. Therefore, $\frac{1}{2}\|\mathbf{p} - \mathbf{q}\|_1 = \frac{1}{2}(|0 - 1| + |1 - 0|) = 1$.

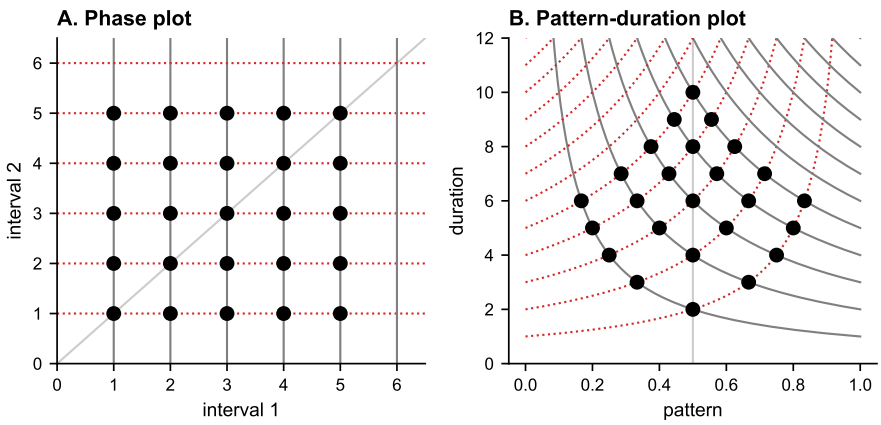

Figure 1: A coordinate transform turns the phase space into a pattern-duration space.

Conversely, multiplying a pattern by its duration moves back to the original space. Formally, the function Φ takes care of all this:

$$\Phi(\mathbf{x}) = \left(\frac{\mathbf{x}}{\|\mathbf{x}\|_1}, \ \|\mathbf{x}\|_1 \right), \qquad \Phi^{-1}(\mathbf{p}, d) = d \cdot \mathbf{p} \tag{4}$$

Geometrically, the pattern-duration decomposition described here can be understood as the *taxicab equivalent of spherical coordinates*. A radial-angular decomposition (the more general term for spherical coordinates) locates a point using a radius and a direction. The direction is specified by several angles that determine a point on the unit ball. Since segments are non-negative sequences, we are only concerned with the non-negative part of $\mathbb{R}^n$ (i.e., the non-negative orthant). In taxicab geometry, the unit ball on that part of the space coincides precisely with a simplex. And so a pattern—a point on the simplex—specifies a direction as a point on the unit ball, while the radius under the L_1 norm equals the duration of a segment. In short, the pattern gives a direction and the duration is the radius.

Knowing how to move between segment space and pattern-duration space, we can pull the taxicab distance from the segment space to define the *pattern-duration distance* d_{pd} on the pattern-duration space:

$$d_{pd}\Big((\mathbf{p}, d), (\mathbf{q}, e) \Big) = \|d\mathbf{p} - e\mathbf{q}\|_1 \tag{5}$$

This distance plays well with the underlying distances. If two pattern-duration pairs share their patterns but not their durations, their distance is the absolute duration difference:

$$\|d\mathbf{p} - e\mathbf{p}\|_1 = |d - e| \cdot \|\mathbf{p}\|_1 = |d - e|, \tag{6}$$

where we use that patterns are normalized, so that $\|\mathbf{p}\|_1 = 1$. Conversely, if two pairs have the same duration e, but different patterns, their distance reduces to

the distance between the patterns multiplied by twice the duration:

$$\|d\mathbf{p} - d\mathbf{q}\|_1 = d \cdot \|\mathbf{p} - \mathbf{q}\|_1 = 2d \cdot d_p(\mathbf{p}, \mathbf{q}). \tag{7}$$

The factor two compensates for the fact that the total variation distance is half the taxicab distance.

2.3 Anisochrony and the nPVI

In this section we define a measure of *anisochrony* and show how it relates to the normalized pairwise variability index (nPVI). First, some notation and properties. Let *centroid* of a simplex be written as $\mathbf{c} = (\frac{1}{n}, \ldots, \frac{1}{n})$. The distance from any corner $\mathbf{p}$ to the centroid is $(n-1)/n$. To see this, note that $\mathbf{p}$ contains only zeros and a single one. It follows that the distance to the centroid is

$$\tfrac{1}{2}\Big((n-1) \cdot |0 - \tfrac{1}{n}| + |1 - \tfrac{1}{n}|\Big) = \tfrac{1}{2}\Big(\frac{n-1}{n} + \frac{n-1}{n}\Big) = \frac{n-1}{n}. \tag{8}$$

We can use this to define the anisochrony of a pattern as the normalized distance from isochrony (i.e., the centroid). Although we define it for patterns, we can naturally extend the definition to segments.

Definition 3. The *anisochrony* of a pattern $\mathbf{p}$ is the normalized distance from isochrony, represented by the centroid $\mathbf{c}$:

$$\alpha(\mathbf{p}) \;:=\; \frac{n}{n-1} \cdot d_p(\mathbf{p}, \mathbf{c}) \;=\; \frac{n}{2(n-1)} \sum_{i=1}^{n} |p_i - \tfrac{1}{n}| \tag{9}$$

The anisochrony of a segment $\mathbf{x}$ is the anisochrony of its pattern $\mathbf{x}/\|\mathbf{x}\|_1$.

For patterns of length $n = 2$ this definition can be simplified. To that end, note that when two positive numbers a and b sum to one, we have

$$|a - \tfrac{1}{2}| = |b - \tfrac{1}{2}| \qquad \text{and} \qquad |a - b| = |a - \tfrac{1}{2}| + |b - \tfrac{1}{2}|. \tag{10}$$

The first property follows directly from $a = b - 1$. For the second one, write $|a - b| = |a - (1 - a)| = |2a - 1| = 2|a - 1/2| = |a - 1/2| + |b - 1/2|$, using the first property in the last step. The second equality proves the following result, when noting that the normalizing constant in the definition of anisochrony disappears when $n = 2$.

Theorem 1. *The anisochrony of a pattern of length 2 is* $\alpha((p_1, p_2)) = |p_1 - p_2|$.

Now we can turn to the normalized pairwise variability index a sequence of intervals $(i_1, \ldots, i_K)$, which is defined as

$$\text{nPVI} = \frac{200}{K-1} \sum_{k=1}^{K-1} \Big|\frac{i_k - i_{k+1}}{i_k + i_{k+1}}\Big|. \tag{11}$$

The previous result almost immediately shows that the nPVI is proportional to the average anisochrony.

Theorem 2. *The nPVI is 200 times the average anisochrony of length-2 segments.*

Proof. Consider intervals $i_1, \ldots i_K$. Theorem 1 shows that the anisochrony of the k-th pattern $\mathbf{p}_k$ is

$$\alpha(\mathbf{p}_k) = \left|\frac{i_k - i_{k+1}}{i_k + i_{k+1}}\right|, \quad \text{where } \mathbf{p}_k = \left(\frac{i_k}{i_k + i_{k+1}}, \frac{i_{k+1}}{i_k + i_{k+1}}\right), \tag{12}$$

which is summand in the definition of the nPVI. It follows that the average anisochrony $\frac{1}{K-1}\sum_k \alpha(\mathbf{p}_k)$ equals $1/200 \times$ nPVI. □

3 Visualization

The main text compares visualizations of three types of synthetic rhythmic data. These supplements include some additional plots for three more synthetic datasets. Interval distributions of all datasets are shown in figure 3.

- **Uniform quantal**. Integers k were sampled uniformly between 1 and 11. Gaussian noise $\epsilon \sim \text{Normal}(0, 0.1)$ was added, and the result scaled by a quantum of 0.2 to get intervals $i = 0.2(k + \epsilon)$.
- **Geometric quantal**. Integers k were sampled from a geometric distribution with paramter $p = 0.5$ to favour small integers. This results in intervals $i = 0.2(k + \epsilon)$ with $\epsilon \sim \text{Normal}(0, 0.1)$ as before.
- **Uniform**. Intervals were sampled uniformly between 0.2 and 2.
- **Normal**. Intervals were sampled from a $\text{Normal}(1, 0.5)$ distribution, truncated at 0, so only positive samples were retained.
- **Isochrony**. Idem, but with smaller varaiance: using $\text{Normal}(0, 0.05)$.
- **Repeated**. The template $(1, 1, 2, 1, 2, 2, 1, 2, 1, 1, 2, 1, 2, 2, 1, 2, 3, 3, 2, 4)$ of integer intervals was repeated (see figure 2), Gaussian noise $\epsilon \sim \text{Normal}(0, 0.1)$ was added and before, and the results was scaled by a quantum of 0.5 to get intervals $i = 0.5(k + \epsilon)$.

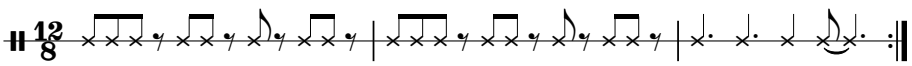

Figure 2: The repeated dataset consists of repetitions of this rhythm, with some noise added. The reader may recognize the first two bars as the backbone of Steve Reichs *Clapping Music*.

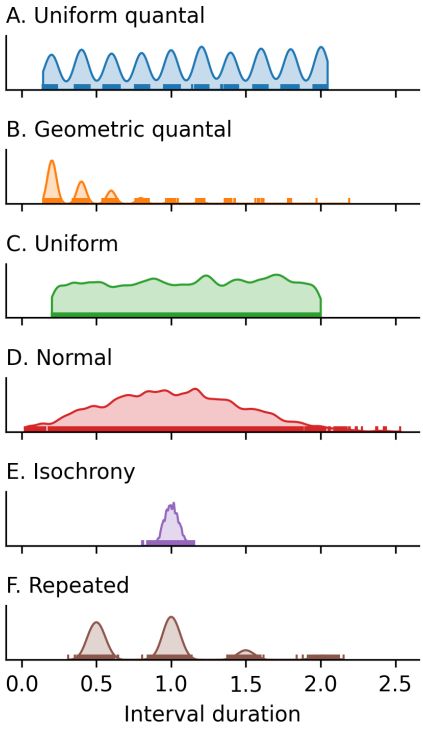

Figure 3: Interval distributions for all synthetic datasets.

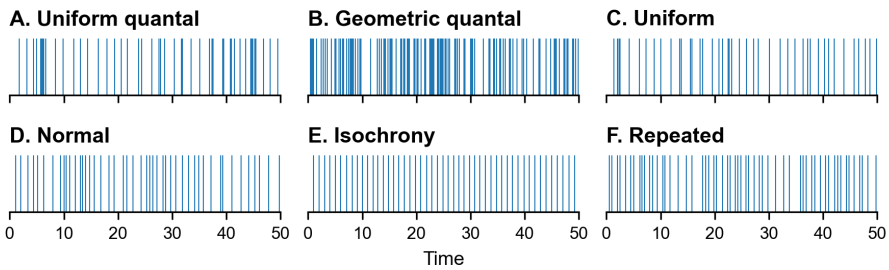

Figure 4: Event plots of synthetic rhythmic data in which each event is marked with a vertical line.

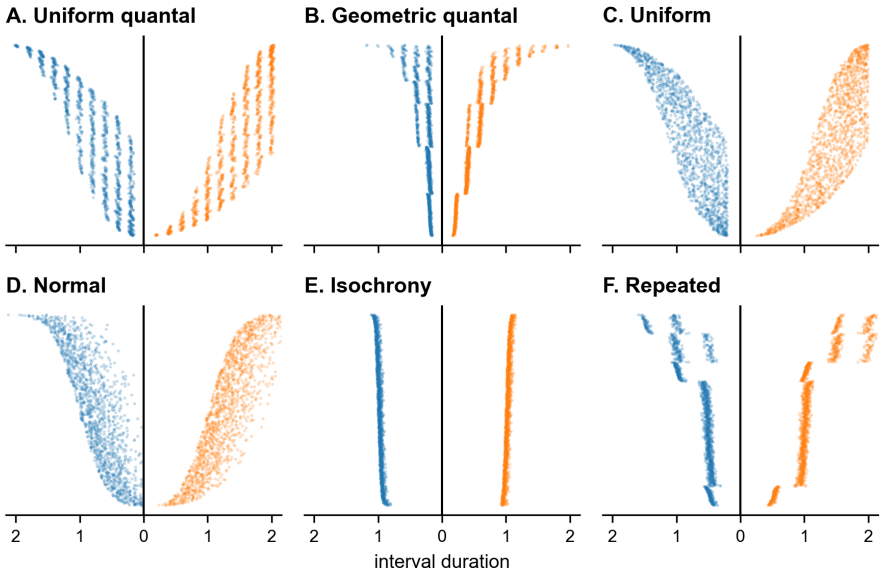

Figure 5: Raster plots for synthetic rhythmic data. Every segment is plotted using two points: the shorter of the two intervals on the left in blue and the longer one on the right in orange. Segments are sorted vertically by their duration, with shorter segments at the bottom.

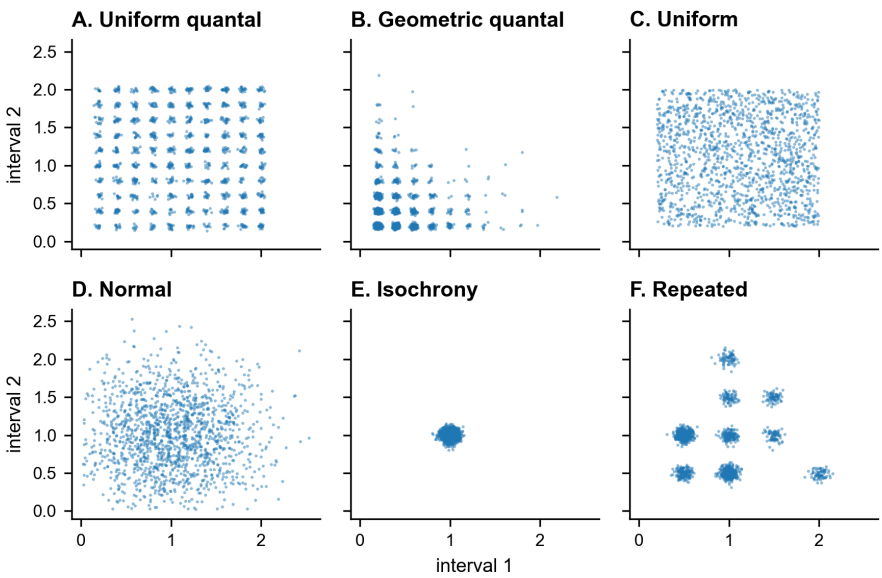

Figure 6: Scatter phase plots of synthetic rhythmic data where each segment is shown as a point, with the duration of the first interval horizontally and the duration of the second interval vertically.

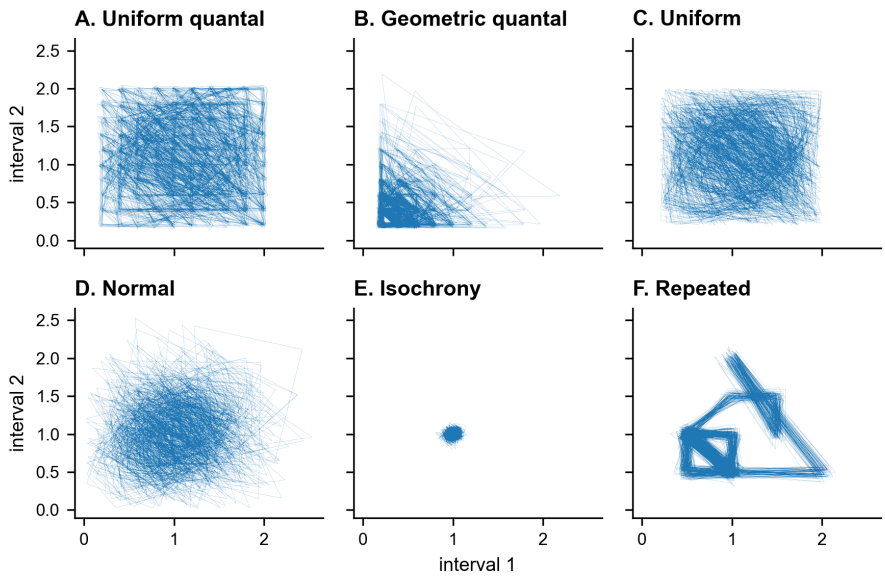

Figure 7: Line pase plots of synthetic rhythmic data. Similar to figure 6, but now with a trajectory connecting successive segments. The trajectory can obfuscate the clustering structure, but in data with recurrent regularities it produces distinct geometric shapes (e.g. subfigure F).

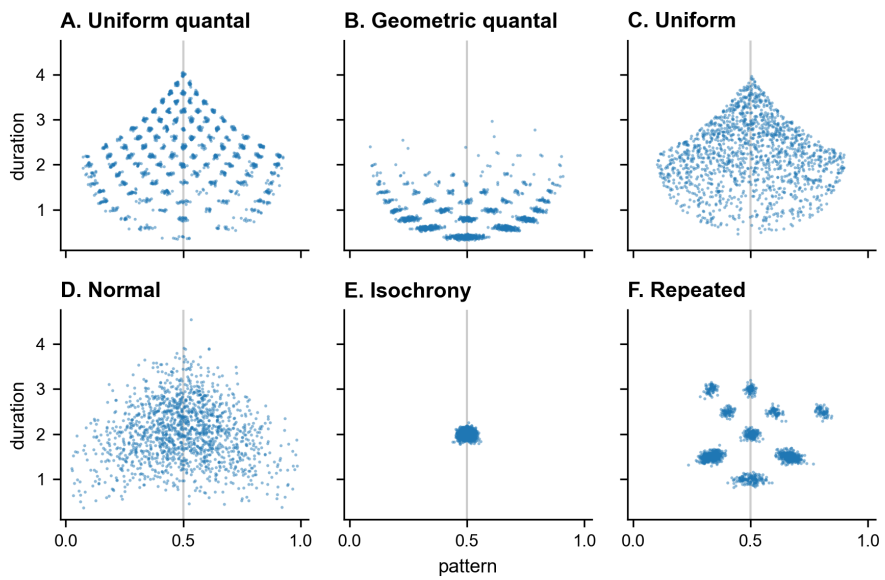

Figure 8: Pattern-duration plots for synthetic rhythmic data.

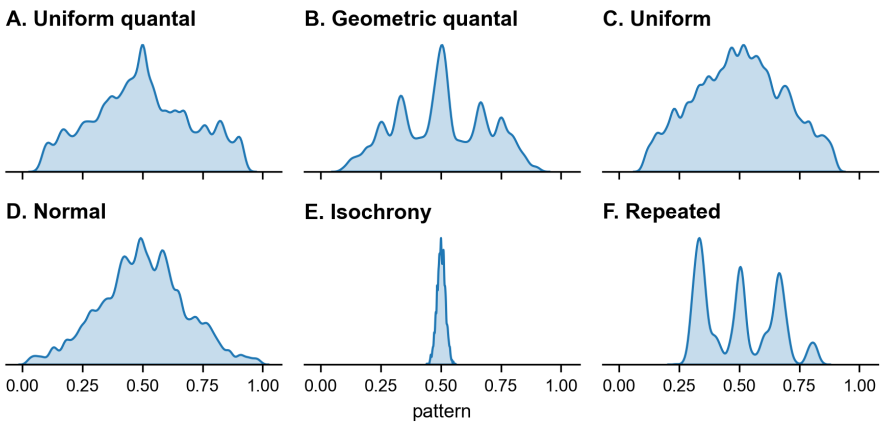

Figure 9: Ratio plots for synthetic rhythmic data.